

VKB-Training: Epistemically Categorized Training Data for Hallucination Reduction in Large Language Models

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Abstract

Large language models hallucinate because their training data carries no epistemic metadata: facts, hypotheses, value judgments, and acknowledged unknowns occupy the same embedding space with identical weight. We propose **VKB-Training** (Verified Knowledge Base Training), a data-centric approach that assigns each training sample a five-category epistemic tag (FACT, MODEL, VALUE, HYPOTHESIS, BLINDSPOT), a calibrated confidence score, and a provenance chain. We introduce a four-stage hybrid annotation pipeline — AI triangulation (where inter-model disagreement signals normative content), human sampling with axiom extraction, expert calibration with reputation weighting, and formal logical consistency filters — alongside three training mechanisms: (1) a confidence-weighted loss function that down-weights uncertain claims and actively unlearns falsified content; (2) provenance-aware attention biasing verified tokens over unverified ones; and (3) a BLINDSPOT training objective that maximizes output entropy when the training data explicitly marks knowledge gaps. We provide formal definitions, a concrete annotation taxonomy, and a falsifiable experimental protocol with pre-specified success thresholds. VKB-Training reframes hallucination as a data problem, not a model problem, and proposes a solution at the layer where the problem originates.

1 Introduction

Current LLM training collapses all knowledge into a single distributional representation:

$$D_{\text{train}} = \{\text{Wikipedia, Reddit, Books, ArXiv, } \dots\} \quad (1)$$

No distinction is made between empirically verified facts, theoretical interpretations, normative value judgments, untested hypotheses, and deliberate misinformation. All enter the same embedding space with identical training weight.

This architectural choice has a direct consequence: the model cannot distinguish $P(\text{fact} \mid s)$ from $P(\text{opinion} \mid s)$, producing what the field calls “hallucination”—confident assertions about uncertain or false content [Ji et al.(2023), Huang et al.(2023)].

Existing mitigations operate *after* the damage is done. RLHF [Ouyang et al.(2022)] fine-tunes on human preferences but inherits Goodhart’s Law: the model optimizes for appearing truthful, not for being truthful. Retrieval-augmented generation [Lewis et al.(2020)] grounds outputs in external documents but cannot assess the epistemic status of those documents. Constitutional

AI [Bai et al.(2022)] filters outputs against rules but does not address the training data that generates them.

We propose a **data-centric** solution: modify the training data itself to carry epistemic metadata, and modify the training procedure to respect that metadata. If the model learns *what it knows*, *how confidently it knows it*, and *where it does not know*—at training time, not inference time—the structural cause of hallucination is addressed at its origin.

2 VKB Taxonomy

2.1 Epistemic Categories

We define five mutually exclusive categories for training data:

Table 1: VKB epistemic taxonomy.

Category	Definition	Example
FACT	Empirically verifiable, independently confirmed	“Water boils at 100°C at 1 atm”
MODEL	Theoretical interpretation or framework	“Inflation is driven by monetary supply”
VALUE	Normative judgment or ethical claim	“Privacy is a fundamental right”
HYPOTHESIS	Testable but unconfirmed claim	“Drug X may reduce mortality by 20%”
BLINDSPOT	Explicitly marked knowledge gap	“Causal mechanism unknown”

2.2 Training Sample Structure

Each VKB training sample is a five-tuple:

$$x_i = \langle s_i, t_i, \sigma_i, p_i, \tau_i \rangle \quad (2)$$

where $s_i \in \mathcal{L}$ is the text, $t_i \in \{\text{FACT}, \text{MODEL}, \text{VALUE}, \text{HYPOTHESIS}, \text{BLINDSPOT}\}$ is the epistemic tag, $\sigma_i \in [0, 1]$ is the confidence score, p_i is the provenance chain (source identifiers, peer review status, evidence links), and τ_i is a timestamp for temporal validity.

2.3 Annotation Strategy

Full annotation of internet-scale corpora is infeasible. We propose a four-stage hybrid pipeline that leverages the complementary strengths of AI, crowd, experts, and formal logic to separate empirically verifiable claims (FACT) from interpretive or normative content (MODEL, VALUE) — a distinction we term the “Caesar/God boundary” following the SVE ontological framework [Kovnatsky(2025b)].

Stage 1 — AI triangulation. Multiple independent LLMs classify the same sample. Where models agree on category assignment, the consensus tag is accepted with high confidence. Where models *disagree*, the disagreement itself is diagnostic: persistent inter-model divergence on a given sample is a strong signal that it contains normative or interpretive content (i.e., lies closer to the VALUE/MODEL boundary than to FACT). Formally: let $t_i^{(k)}$ be the tag assigned by model k to sample i . Define the disagreement index $D_i = 1 - \max_t \frac{|\{k: t_i^{(k)} = t\}|}{K}$, where K is the number of models. High D_i triggers escalation to Stage 2.

Stage 2 — Human sampling and axiom extraction. Domain-literate annotators review a curated subset ($\sim 10\%$ of training data), with priority given to high- D_i samples from Stage 1.

Persistent disagreements between annotators are tagged BLINDSPOT. An alternative pathway uses the Epistemological Boxing Protocol (EBP) [Kovnatsky(2025)]: a claim is submitted to structured adversarial dialogue, producing an Integrity Score and epistemic category. Annotator decisions are collected to (a) train a lightweight neural classifier for propagation to the remaining 90%, and (b) extract recurrent axioms — fundamental principles that annotators invoke to resolve borderline cases. These axioms, once identified, become reusable decision rules.

Stage 3 — Expert calibration with reputation weighting. Following the Distributed IVM architecture [Kovnatsky(2025b)], expert reviewers carry calibrated reputation scores based on their track record (accuracy on previously resolved samples). Each expert’s tag is weighted by their reputation score, implementing a formalized version of Galton’s ox-weighing insight: collective expert judgment outperforms any individual, provided judgments are independent and weighted by demonstrated competence. Formally: $\sigma_i^{\text{expert}} = \sum_e w_e \cdot \sigma_{e,i}$, where w_e is the reputation weight of expert e .

Stage 4 — Logical consistency filters. Final annotations pass through formal consistency checks inspired by the Cauchy-Gödel-Socrates (CGS) Method [Kovnatsky(2026b)]: (a) internal contradiction detection (does the sample contradict other FACT-tagged samples in the corpus?), (b) symmetry verification (would the same claim, with subjects swapped, receive the same tag?), and (c) semantic interval bisection (can the claim be decomposed into a factual component and a normative component?). Samples failing consistency checks are re-routed to Stage 2.

A key hypothesis: even 10% expert-verified annotation, propagated through confidence-weighted loss, can produce measurable hallucination reduction — because the four-stage pipeline ensures that the verified 10% is maximally informative (high-disagreement, boundary cases resolved by experts and logic, not random samples).

3 Training Mechanisms

3.1 Confidence-Weighted Loss

Standard cross-entropy loss treats all tokens equally:

$$\mathcal{L}_{\text{standard}} = - \sum_{i=1}^N \log P_{\theta}(s_i \mid \text{context}_i) \quad (3)$$

VKB-weighted loss incorporates epistemic metadata:

$$\mathcal{L}_{\text{VKB}} = - \sum_{i=1}^N \sigma_i \cdot \log P_{\theta}(s_i \mid \text{context}_i, t_i) \quad (4)$$

Three properties follow directly:

1. Low-confidence samples ($\sigma < 0.5$) contribute proportionally less — the model learns to underweight uncertain content.
2. Falsified content ($\sigma = 0$, tagged after retraction or correction) contributes zero loss — effectively unlearned.
3. Type conditioning (t_i) enables the model to learn that the *same* surface form (“X causes Y”) has different epistemic weight when tagged FACT vs. HYPOTHESIS.

The loss structure is analogous to the vectorial purification process introduced in the Epistemological Boxing Protocol [Kovnatsky(2025)], where a thesis vector is iteratively refined by subtracting identified error vectors: $\vec{v}^{(j+1)} = \vec{v}^{(j)} - \vec{\epsilon}_j$. In VKB-Training, the confidence weight σ_i plays the role of the error magnitude: low-confidence samples are “subtracted” from the model’s certainty landscape during training, achieving the same iterative purification at the level of the training corpus rather than a single argument.

3.2 Provenance-Aware Attention

Standard transformer attention [Vaswani et al.(2017)]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \quad (5)$$

We add a provenance bias term:

$$\text{Attention}_{\text{prov}}(Q, K, V, P) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + \alpha \cdot P\right) V \quad (6)$$

where P_{ij} is the provenance score for token j :

$$P_{ij} = \begin{cases} +\log(\sigma_j) & \text{if token } j \text{ from verified source} \\ -\infty & \text{if token } j \text{ from falsified source} \\ 0 & \text{if unverified} \end{cases} \quad (7)$$

The model preferentially attends to verified information and is structurally prevented from attending to falsified content.

3.3 BlindSpot Training: Learning What You Don’t Know

Current LLMs hallucinate most when they are uncertain [Kadavath et al.(2022)]. We propose an explicit training signal for uncertainty.

For samples tagged BLINDSPOT, we *maximize* output entropy rather than minimizing cross-entropy:

$$\mathcal{L}_{\text{blind}} = - \sum_{i: |\text{BlindSpots}(s_i)| > 0} H(P_\theta(s_i \mid \text{context})) \quad (8)$$

This trains the model to produce high-entropy (uncertain) outputs precisely where the training data explicitly marks knowledge gaps. The model learns: “when the evidence is contested or absent, express uncertainty rather than confabulate.”

3.4 Epistemic Embedding Decomposition

We decompose the embedding space $\mathbb{R}^d$ into orthogonal epistemic subspaces:

$$\mathbb{R}^d = \mathcal{F} \oplus \mathcal{M} \oplus \mathcal{V} \oplus \mathcal{U} \quad (9)$$

where $\mathcal{F}$ (factual), $\mathcal{M}$ (model/interpretive), $\mathcal{V}$ (value), and $\mathcal{U}$ (uncertainty) are learned subspaces. Projection operators $\Pi_{\mathcal{F}}$, $\Pi_{\mathcal{M}}$, $\Pi_{\mathcal{V}}$, $\Pi_{\mathcal{U}}$ extract the epistemic component of any embedding, enabling the model to decompose a mixed statement (e.g., “GDP grew 3%, suggesting the policy worked”) into its factual and interpretive parts.

4 Proposed Experiment

4.1 Setup

Base model: Open-weight 7B-parameter model (e.g., Llama 3 or Mistral).

Training data: A curated subset of 500K samples from Wikipedia, PubMed abstracts, and news articles, annotated using the four-stage VKB pipeline (Stage 1 AI triangulation, Stage 2 human verification for high-disagreement samples, Stage 3 expert calibration, Stage 4 logical consistency filters).

Conditions:

- **Baseline:** Standard fine-tuning on the same 500K samples without epistemic tags.
- **VKB-Partial:** Fine-tuning with VKB loss (Eq. 4) on the 10% expert-verified subset; standard loss on the remainder.
- **VKB-Full:** Fine-tuning with VKB loss on all 500K samples (Stage 1 automated tags + Stage 2/3 corrections).
- **VKB-Full+Attn:** VKB-Full plus provenance-aware attention (Eq. 6).
- **VKB-Full+Attn+Blind:** All mechanisms including BlindSpot training (Eq. 8).

4.2 Evaluation

Benchmarks: TruthfulQA [Lin et al.(2022)], HaluEval [Li et al.(2023)], FActScore [Min et al.(2023)].

Primary metric: Reduction in *confident-but-wrong* answers — responses where the model assigns high probability (> 0.8) to factually incorrect statements.

Secondary metrics: (a) Calibration error (ECE) across epistemic categories. (b) Selective abstention rate: does the model abstain more on BLINDSPOT topics? (c) Factual accuracy on FACT-tagged domains.

4.3 Pre-Specified Success Thresholds

Table 2: Falsification criteria.

Outcome	Interpretation
$< 15\%$ hallucination reduction	Hypothesis weakened; epistemic tags insufficient
15–30% reduction	Partial confirmation; data-centric approach viable
30–50% reduction	Strong confirmation; publishable result
$> 50\%$ reduction	Paradigm shift; VKB as industry standard candidate

5 Related Work

Epistemic tagging has been explored in knowledge graphs [Hogan et al.(2021)] but not applied to LLM training data. Confidence calibration has been studied for classification [Guo et al.(2017)] but not for generative models with provenance metadata. Data-centric AI [Zha et al.(2023)] emphasizes data quality but has not proposed a formal epistemic taxonomy. RAG [Lewis et al.(2020)] addresses hallucination at inference time; VKB addresses it at training time — the two are complementary.

The use of multi-model disagreement as a signal for epistemic status has precedent in ensemble methods and inter-annotator agreement studies, but has not been applied as a *classifier for the Fact/Value boundary* in training data. The reputation-weighted expert system (Stage 3) extends the collective intelligence formalization of [Kovnatsky(2025b)], which revisits Galton’s ox-weighing insight under conditions of known expert calibration. The logical consistency filters (Stage 4) draw on the CGS Method [Kovnatsky(2026b)], which provides formal protocols for detecting contradictions between factual claims and normative constraints within language model outputs.

The BlindSpot mechanism relates to “I don’t know” training [Kadavath et al.(2022)] but differs critically: VKB encodes uncertainty *in the training data itself*, making epistemic humility a first-class training signal. The concept of “virtuous concession”—treating acknowledgment of ignorance as progress—was formalized in the Epistemological Boxing Protocol [Kovnatsky(2025)] and directly informs VKB’s BLINDSPOT category.

6 Discussion

Annotation cost and efficiency. Full VKB annotation at internet scale is infeasible. The four-stage pipeline (Section 2.3) addresses this by concentrating human effort where it matters most: Stage 1 AI triangulation filters out easy cases automatically; only high-disagreement samples ($D_i > 0.3$) are escalated to human review. The EBP-based adversarial pathway and CGS logical filters [Kovnatsky(2026b)] further reduce the volume requiring expert attention. Our hypothesis: 10% expert-verified annotation, selected by disagreement-maximizing sampling, outperforms 10% random annotation by a significant margin.

Taxonomy granularity. Five categories may be too coarse or too fine. Inter-annotator agreement on boundaries — particularly MODEL vs. HYPOTHESIS — is an empirical question addressable in the proposed experiment.

Provenance bootstrapping. Initial provenance scores can be bootstrapped from existing trust signals: journal retraction databases, Wikidata confidence ratings, and institutional source rankings. Imperfect but available now.

Complementarity. VKB-Training does not replace RLHF, RAG, or Constitutional AI. It addresses the training data layer and can be combined with all post-training methods. Gains are expected to be additive.

Broader implications. If epistemic categorization reduces hallucination significantly, the hallucination problem is fundamentally a *data provenance* problem, not an architecture problem — redirecting the field toward training data quality rather than model scale.

7 Ethical Data Sourcing: Author Revenue Sharing

Note: This section is included in the Zenodo preprint for completeness. It will be omitted from the workshop submission to maintain technical focus.

VKB-Training depends on high-quality epistemic annotations. These annotations require human expertise — domain researchers who understand whether a claim is a FACT, a MODEL, or a HYPOTHESIS. Current industry practice treats training data as a free resource extracted from authors without compensation or consent, producing an adversarial dynamic (ongoing litigation: NYT v. OpenAI, Authors Guild v. OpenAI, Getty v. Stability AI).

We propose an alternative: **authors who permit VKB annotation of their work receive a revenue share from models trained on that data.** The structure:

- **10–25%** of attributable revenue during the competitive scaling phase (years 1–5), when margins are thin and market position is being established.
- **25–50%** at operational maturity, when the VKB dataset has become a differentiated asset. The exact percentage is determined by community vote (Veche), not unilaterally.
- **Requests for increases beyond 50%** are accepted and decided collectively. No cap is promised; no specific figure above 50% is guaranteed.

In exchange, authors consent to epistemic categorization of their work (a paper may be tagged HYPOTHESIS rather than FACT — intellectual honesty requires accepting this). Authors receive full financial transparency (all revenue data is auditable) and are invited to SVE Family membership with voting rights on revenue allocation and framework governance.

This creates the first *ethically sourced, epistemically categorized training dataset* — a competitive advantage that cannot be replicated by organizations whose business model depends on uncompensated data extraction. The ethical sourcing itself becomes the moat.

The mechanism is governed by SVE Meta-License v4.0, Appendix B (Ethical Partnership Model). Violations — using VKB-annotated data without paying the agreed share — trigger standard SVE enforcement (Appendix E, Attorney Incentive Program).

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VKB-Training was first described as part of the CogOS framework [Kovnatsky(2026)]. The vectorial purification formalism and adversarial annotation originate from the Epistemological Boxing Protocol [Kovnatsky(2025)]. The expert reputation-weighting system and DAG-based knowledge verification derive from S.V.E. XI (The Ox’s Weights) [Kovnatsky(2025b)]. The logical consistency filters are based on the CGS Method [Kovnatsky(2026b)]. The present paper integrates these components into a standalone, empirically testable training proposal.

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Repository: <https://github.com/skovnats/SVE-Systemic-Verification-Engineering>

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